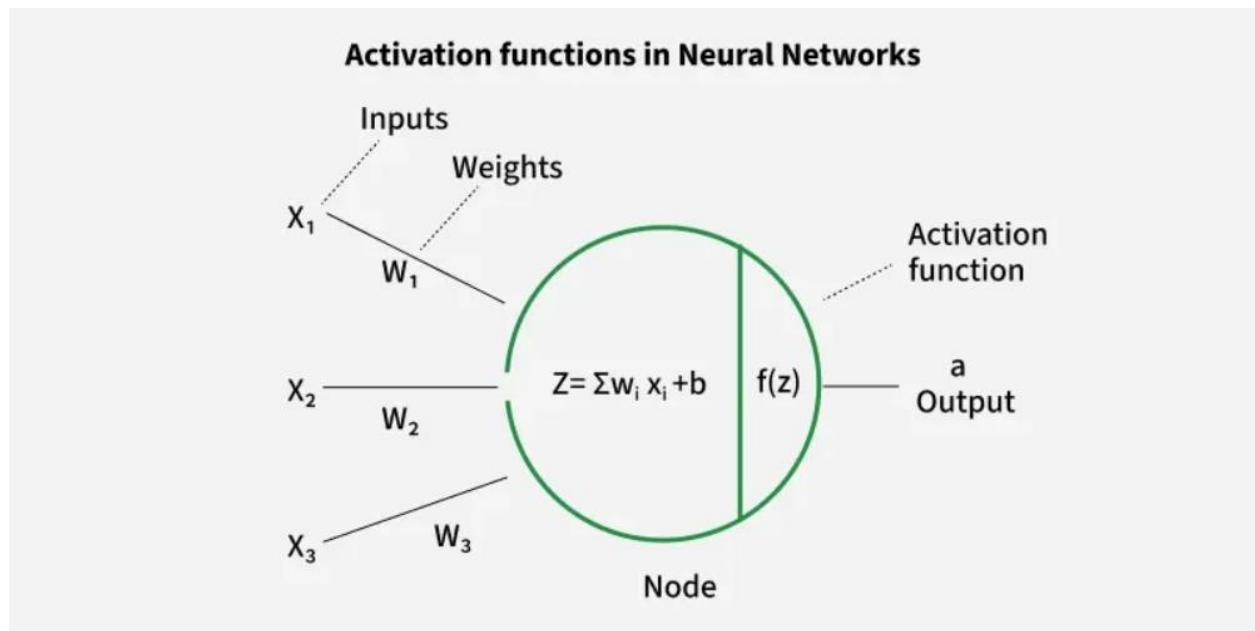


Activation functions in Neural Networks

An activation function in a neural network is a mathematical function applied to the output of a neuron. It introduces non-linearity, enabling the model to learn and represent complex data patterns. Without it, even a deep neural network would behave like a simple linear regression model.

Activation functions decide whether a neuron should be activated based on the weighted sum of inputs and a bias term. They also make backpropagation possible by providing gradients for weight updates.



Why Non-Linearity is Important

- Real-world data is rarely linearly separable.
- Non-linear functions allow neural networks to form **curved decision boundaries**, making them capable of handling complex patterns (e.g., classifying apples vs. bananas under varying colors and shapes).
- They ensure networks can model advanced problems like image recognition, NLP and speech processing.

Impact of Activation Functions on Model Performance

1. **Convergence Speed:** Functions like ReLU allow faster training by avoiding the vanishing gradient problem while Sigmoid and Tanh can slow down convergence in deep networks.
2. **Gradient Flow:** Activation functions like ReLU ensure better gradient flow, helping deeper layers learn effectively. In contrast Sigmoid can lead to small gradients, hindering learning in deep layers.
3. **Model Complexity:** Activation functions like Softmax allow the model to handle complex multi-class problems, whereas simpler functions like ReLU or Leaky ReLU are used for basic layers.